^{177}Lu - Comments on evaluation of decay data

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The initial evaluation was completed in February 2002 by F.G. Kondev from ANL. A complete re-evaluation was carried out in April 2015 by M.A. Kellett from CEA-LNHB (2016Ke06), taking into account new results from the radionuclide metrology community presented at the ICRM2015 conference and eventually published in 2016. The present evaluation is an update based on available literature by mid October 2025, with changes mainly to the Q-value and the half-lives.

Evaluation Procedures

The *Limitation of Relative Statistical Weights Method* (LWM) (1988WoZO) as implemented in the LWEIGHT program has been used throughout this evaluation.

1. Decay Scheme

The decay scheme for ^{177}Lu is approximately based on the evaluation of Kondev (2003Ko33 and in 2004BeZQ), however the multipolarities have been completely re-evaluated using the BrIccMixing code and the β transition probabilities have now been determined from the γ transition probability for each level. The ground state has been assigned $J^{\pi} = 7/2^{+}$ and $7/2^{+}[404]$ ($g_{7/2}$) in Nilsson configuration. It decays via β^{-} emission ($P_{\beta-} = 100\%$) to levels of the stable ^{177}Hf daughter isotope. While the decay branches to the ^{177}Hf ground state ($J^{\pi} = 7/2^{-}$) as well as to the 112.95005-keV ($J^{\pi} = 9/2^{-}$) and 321.3159-keV ($J^{\pi} = 9/2^{+}$) levels are well established, there is some ambiguity in the literature regarding the direct β^{-}-decay feeding into the ($J^{\pi} = 11/2^{-}$) level at 249.6742 keV. In this evaluation, the feeding to this level has a 100% uncertainty associated with it. Nonetheless, a consistent decay scheme has been established with a total decay energy release of 496.8 (19) keV, compared to the adopted Q-value of 496.8 (8) keV from 2021Wa16.

2. Nuclear Data

2.1 Half-lives

The half-life of the ^{177}Lu ground state has been measured by several authors and the results are summarised in Table 1. In all cases, except two, the source was prepared using the ^{176}Lu(n,γ) reaction, where a three-quasiparticle isomer ($K^{\pi} = 23/2^{-}$ with an excitation energy of 970 keV), with a significantly longer half-life ($T_{1/2}$ = 160.44 (6) d), is also produced. Since the isomer de-excites partially via gamma emission (P_{γ} = 21.4 (8) %), its half-life and relative population need to be taken into account when determining the $T_{1/2}$ for the ground state. In two studies, from 2016Dr15 and 2017MeZZ, the source ampoule was provided by the same manufacturer via the ^{176}Yb(n,γ)^{177}Yb reaction followed by β-decay to ^{177}Lu, and no ^{177m}Lu activity was identified. It is noteworthy that the results from 2016Dr15 and 2023Ma26 are from the same laboratory. However, different sources and measurement techniques were employed. Therefore, the evaluators chose to consider them as two independent values.

The recommended value for the ^{177}Lu ground state half-life is $T_{1/2}$ = 6.6443 (9) d, which is the weighted mean of the eight values in bold in Table 1. Two of the recent half-life measurements reported in 2012Re25 and 2016Lu16 were rejected due to the Chauvenet criterion, and the half-lives given in 1958Be41, 1960Sc19, 1972Em01, 1990Ab02 and 2008Ca13 were excluded from this analysis since the authors did not consider the effect of the ^{177m}Lu isomer ($T_{1/2}$ = 160.44 d).

The value of 2017FeZZ has the highest relative statistical weight of 64%. It should be noted that a systematic uncertainty in the recommended $T_{1/2}$ value for the ^{177}Lu ground state could exist due to possible differences in the half-life values of ^{177m}Lu and its activity used in 1982La25, 2004Sc04, 2011Po07, 2012Fi12, 2012Ko24 and 2023Ma26.

Table 1 – Measured and recommended values for the ^{177}Lu ground state half-life.

Reference	$T_{1/2}$ (d)	Comments
1958Be41	6.75 (5) *	Correction for $T_{1/2}$ (^{177m}Lu) neglected
1960Sc19	6.74 (4) *	Correction for $T_{1/2}$ (^{177m}Lu) neglected
1972Em01	6.71 (1) *	Correction for $T_{1/2}$ (^{177m}Lu) neglected
1982La25	**6.645 (30)**	$T_{1/2}$(^{177m}Lu) = 159.5 (7) d was used in the fitting procedure.
1990Ab02	6.7479 (7) *	Correction for $T_{1/2}$ (^{177m}Lu) neglected – measured for 24 d.
2001Zi01	6.65 (1)	Corrections for $T_{1/2}$ (^{177m}Lu) have been applied, but the value has not been reported. Mean value from IC and LSC measurements. Superseded by 2002Un02.
2001Sc23	6.646 (5)	IC measurement – $T_{1/2}$ (^{177m}Lu) = 160.4 d was used in the fitting procedure. Superseded by 2004Sc04.
2002Un02	6.64 (1)	Replaces 2001Zi01 and is superseded by 2012Fi12.
2004Sc04	**6.6475 (20)**	Re-analysis of PTB IC data and supersedes 2001Sc23.
2008Ca13	6.709 (1) *	Correction for $T_{1/2}$ (^{177m}Lu) neglected – measured for 40 d.
2011Po07	**6.6465 (50)**	Mean of two results, one from IC 6.646 (7), the other LSC 6.647 (7).
2012Fi12	**6.640 (10)**	Replaces 2002Un02.
2012Ko24	**6.639 (9)**	Analysis of PTB LSC data from 2004 never before published. Independent of 2001Sc23 which were IC measurements.
2012Re25	6.681 (6) @	Measurement over 25 days (16 data points), but not clear how ^{177m}Lu activity taken into account.
2016Lu16	6.660 (17) @	Unweighted mean of two values which use different techniques to correct for the ^{177m}Lu activity.
2016Dr15	**6.645 (3)**	Unweighted mean of three spectrometry and one ionisation chamber measurement. Authors' state that source contains negligible ^{177m}Lu activity.
2017FeZZ	**6.6430 (11)**	Ionisation chamber measurement. Same source manufacturer as in 2016Dr15 and same conclusion about the absence of ^{177m}Lu activity.
2023Ma26	**6.6489 (52)**	Unweighted mean of two measurements (liquid scintillation counting, double and triple coincidences).
Adopted	**6.6443 (9)**	**χ^2 = 0.79, c.f. χ^2_{crit} = 2.64**

* Contributions from the decay of the ^{177m}Lu ($T_{1/2}$ = 160.44 d) isomer have not been taken into account. The value is not used in the analysis.

@ Rejected due to the Chauvenet criterion.

As an example of the influence of the presence of ^{177m}Lu has on the half-life measurement for the ground state, Luca *et al.* (2016Lu16) kindly provided their result prior to correction and two values obtained depending on how the ^{177m}Lu activity is determined:

Correction for $T_{1/2}$ (^{177m}Lu) not included:	6.699 (40)
Correction for $T_{1/2}$ (^{177m}Lu) (γ spectrometry):	6.675 (3)
Correction for $T_{1/2}$ (^{177m}Lu) (ionisation chamber):	6.645 (17)
Unweighted mean of two corrected values:	6.660 (17)

It is interesting to note that the uncorrected value from Luca *et al.* (6.699 (40) d) is significantly higher than the corrected values and approaches the values of the other authors who did not correct for the presence of ^{177m}Lu.

Level half-lives have also been updated based on 2019 ENSDF evaluation (2019Ko26).

112.95005-keV (9/2^{-}) level

Five measurements are available: 0.583 (6) ns from 1992De53; 0.5295 (30) ns from 1996Al20; 0.77 (7) ns from 2011Zh56; 0.545 (6) ns from 2013Do24; and 0.535 (10) ns from 2020Kn04. The value from 2011Zh56 is excluded due to the Chauvenet criterion. The dataset is discrepant (χ^2 = 18.9 compared to χ^2_{crit} = 3.8) and the adopted value of 0.545 (15) ns corresponds to the weighted mean with the expanded uncertainty to include the most precise value from 1996Al20.

249.6742-keV (11/2^{-}) level

Three measurements are available: 107 (11) ps from 1961Ha21; 106 (21) ps from 1972Da35; and 95.0 (21) ps from 2020Kn04. The dataset is consistent (χ^2 = 0.70 compared to χ^2_{crit} = 4.61) and the adopted value of 95.5 (21) ps corresponds to the weighted mean with the minimum experimental uncertainty of the dataset.

321.3159-keV (9/2^{+}) level

Seven measurements are available: 0.63 (5) ns from 1961Ha38; 0.69 (3) ns from 1962Be46; 0.65 (6) ns from 1965Ro17; 0.65 (3) ns from 1968Ra25; 0.67 (3) ns from 1973GoYH; 0.728 (21) ns from 2011Zh56; and 0.792 (17) ns from 2020Kn04. The value from 2020Kn04 is excluded due to the Chauvenet criterion. The dataset is consistent (χ^2 = 1.47 compared to χ^2_{crit} = 3.02) and the adopted value of 0.688 (15) ns corresponds to the weighted mean with the external uncertainty.

2.2 Q value

The Q(β^-) = 496.8 (8) keV is from 2021Wa16. It is in agreement with that of 496.8 (17) keV (1962El02), deduced from the β^--decay endpoint energy to the ^{177}Hf ground state. The total decay energy released in the β^--decay of the ^{177}Lu ground state is calculated using RADLST (1988Bu*) as 496.2 (22) keV, and using Saisinuc (2008DuZX) as 496.8 (19) keV. It agrees very well with the Q(β^-) value that is reported by Wang *et al.* (2021Wa16), thus suggesting that the decay scheme is complete.

2.3 Beta Decay Transitions

The β^- transition endpoint energies were determined from Q(β^-) = 496.8 (8) keV (2021Wa16) and the individual level energies of ^{177}Hf. The latter were deduced using the GTOL code from the adopted γ-ray energies given in Table 3. The β^- transition endpoint energies are in agreement with values measured by 1962El02 and 1955Ma12. The adopted values for the β^- transition probabilities per 100 disintegrations given in Table 2 were determined from the total (photons + conversion electrons) transition probability balances at each level. In general, values deduced in the present evaluation are consistent with those from 2001Sc23, albeit the authors did not report on a direct β^- -decay feeding into the J^π = 11/2⁻ level.

Table 2 – Measured and adopted values for the ^{177}Lu β^--decay transition probabilities.

Reference	$P_{\beta-}$ to J^π = 7/2⁻	$P_{\beta-}$ to J^π = 9/2⁻	$P_{\beta-}$ to J^π = 11/2⁻	$P_{\beta-}$ to J^π = 9/2⁺
2001Sc23	79.3 (5)	9.1 (5)		11.58 (12)
1967Ha09	87.2 (11)	6.0 (8)	0.07 (2)	6.7 (3)
1964Al04	86.3 (13)	7 (1)	0.03 (3)	6.7 (3)
1962El02	90 (4)	2.95 (3)	0.31 (6)	6.72 (25)
1956Wi39	96	1.3	0.2	2.6
1955Ma12	90	3		7
1949Do05	65	17		
Adopted	**79.45 (29)**	**8.99 (28)**	**0.003 (21)**	**11.55 (7)**

There are, however, significant differences with the 1967Ha09, 1964Al04, 1962El02, 1956Wi39, 1955Ma12 and 1949Do05 work, as summarised in Table 2. The log *ft* values were calculated using the BetaShape program (2023Mo21) using the adopted β^- transition probabilities.

2.4 Gamma Transitions and Internal Conversion Coefficients

The measured values for γ-ray transition energies that follow the decay of the ^{177}Lu ground state are presented in Table 3. In most cases, the weighted mean of the set of measured energies has been adopted. Clearly the precise γ-ray energies reported by Matsui *et al.* (1989Ma56) and Deepa *et al.* (2011De07) have the greatest influence on the adopted values. Both measurements were made with a high-precision germanium spectrometer.

Transition multipolarities have been completely re-assessed using the BrIccMixing code, which can use all relevant data types, i.e., those coming from measured electron conversion coefficients, as well as angular correlation data and nuclear orientation experiments (see Section 2.2.2).

The total (photon + conversion electrons) transition probabilities were deduced by multiplying the adopted values for the relative γ-ray intensities (Table 10) by a normalisation factor that was deduced from the values for the absolute intensity per 100 disintegrations of the 208.3661-keV γ-ray (Table 11). The electron conversion coefficients are interpolated values using the BrIcc code v2.3 (2008Ki07) from the tables of Band *et al.* (2002Ba85). The quoted uncertainties also reflect the corresponding uncertainties in the mixing ratio values. Adopted α_K, α_{L1}, α_{L2}, α_{L3}, and α_M values were used as an input for the EMISSION code (2001Sc08).

Table 3 – Measured and adopted values for γ-ray emission energies following β^- decay of ^{177}Lu.

Reference	$\gamma_{1,0}$	$\gamma_{2,1}$	$\gamma_{2,0}$	$\gamma_{3,2}$	$\gamma_{3,1}$	$\gamma_{3,0}$
2012De24	112.951 (1)	136.707 (10)	249.655 (3)	71.642 (5)	208.370 (1)	321.347 (13)
2011De07	112.9468 (2)	136.7242 (37)	249.673 (8)	71.6441 (13)	208.3661 (3)	321.327 (1)
1989Ma56	112.9498 (4)	136.7245 (5)	249.6742 (6)	71.6418 (6)	208.3662 (4)	321.3159 (6)
1981Hn03	112.95 (2)	136.72 (2)	249.7 (5)	71.646	208.35 (2)	321.27 (5)
1967Ha09	112.95 (2)	136.72 (5)	249.65 (6)	71.66 (6)	208.34 (6)	321.32 (12)
1965Ma18	112.952 (2)	136.730 (6)	249.868 (25)	71.646 (2)	208.359 (10)	321.330 (40)
1964Al04	112.97 (2)	136.68 (2)	249.69 (10)	71.64 (2)	208.36 (6)	321.36 (20)
1961We11	112.97 (2)	136.70 (10)	249.70 (10)	71.60 (10)	208.38 (2)	321.34 (3)
1955Ma12	112.965 (20)		250.0 (5)	71.644 (20)	208.362 (20)	321.36 (10)
Adopted	**112.95005 (36)**	**136.7245 (5)**	**249.6742 (6)**	**71.6425 (6)**	**208.3661 (2)**	**321.3159 (6)**

2.4.1 Gamma energy determination

112.95005 (36) keV ($\gamma_{1,0}$)

The complete set of data is discrepant (χ^2 = 5.56 compared to χ^2_{crit} = 2.51) due to the value from 2011De07. The study of 1989Ma56 is much more detailed, showing an uncertainty twice as large. It is difficult to understand why the uncertainty of the 2011De07 value is more than ten times lower than for their value of the $\gamma_{2,1}$-ray, which might indicate a typographical error in the reported uncertainty. The authors and experimental device in 2011De07 and 2012De24 being the same, only the latter value was conserved in the analysis. The resulting dataset is then consistent (χ^2 = 0.68 compared to χ^2_{crit} = 2.64). The adopted value is the weighted mean with the internal uncertainty, with an 83% contribution of the value from 1989Ma56.

136.7245 (5) keV ($\gamma_{2,1}$)

The adopted value is the weighted mean of the complete set of data (χ^2 = 0.67 compared to χ^2_{crit} = 2.81). Owing to its vastly superior precision, the value of 1989Ma56 contributes 97% to this value.

249.6742 (6) keV ($\gamma_{2,0}$)

The adopted value is the weighted mean of a reduced set of data (χ^2 = 0.06 compared to χ^2_{crit} = 3.02). The less precise value of 2012De24 has been removed and the values from 1965Ma18 and 1955Ma12 have been removed as inconsistent owing to the Chauvenet criterion.

71.6425 (6) keV ($\gamma_{3,2}$)

The adopted value is the weighted mean of a reduced set of data (χ^2 = 1.49 compared to χ^2_{crit} = 3.32). The less precise value of 2012De24 and the 1981Hn03 value (which has no uncertainty) have been removed and the values from 1961We11 and 1967Ha09 have been removed as inconsistent owing to the Chauvenet criterion.

208.3661 (2) keV ($\gamma_{3,1}$)

The adopted value is the weighted mean of the complete set of data (χ^2 = 0.27 compared to χ^2_{crit} = 2.64), apart from the less precise value of 2012De24. The weighted mean is effectively the weighted mean of the two precise values from 2011De07 (contributes almost 64%) and 1989Ma56 (almost 36%).

321.3159 (6) keV ($\gamma_{3,0}$)

The dataset is very discrepant (χ^2 = 9.47 compared to χ^2_{crit} = 2.64) and owing to the higher than usual uncertainty (and the discrepant nature) of the two values from Deepa *et al*. (2011De07 and 2012De24), the adopted value is the precise value of 1989Ma56.

2.4.2 Mixing ratio determination

Details about the mixing ratio values for E1+M2 and M1+E2 transitions are given below. The recommended values are those calculated with the BrIccMixing code v2.3b (2015Ki*) based on the input data shown below.

112.95005 keV ($\gamma_{1,0}$)

Values used in the analysis of the mixing ratios are summarised in Table 4 with the recommended value calculated using the BrIccMixing code. The individual $\delta(\gamma_{1,0})$ values listed in column 2 of Table 4 are for illustration only, as many of these were determined by the authors by comparing to theoretical conversion coefficients at the time.

136.7245 keV ($\gamma_{2,1}$)

The adopted mixing ratio of -3.2 (+11-9) is calculated using the BrIccMixing code, with a reduced χ^2 of 3.53, based on the values given in Table 5 of $\delta(\gamma_{2,1})$ = -3.0 (7) from 1974Kr12 and -2.38 (36) from

2012De24. The conversion coefficients determined in 1974Ag01 for this transition are not self-consistent and have not been used.

321.3159 keV ($\gamma_{3,0}$)

Values used in the analysis of the mixing ratios are summarised in Table 6. The unweighted mean value is adopted, but the uncertainty was expanded so that the range includes the most precise value of $\delta(\gamma_{3,0})$ = + 0.17 (1) (1974Kr12). The sign of $\delta(\gamma_{3,0})$ was determined to be positive by 1974Kr12.

208.3661 keV ($\gamma_{3,1}$)

Values used in the analysis of the mixing ratios are given in Table 7. The weighted mean and the internal uncertainty were adopted. The sign of $\delta(\gamma_{3,1})$ is uncertain. It has been reported to be positive by 1974Kr12, but negative by 1961We11 and 1977Ke12.

71.6425 keV ($\gamma_{3,2}$)

Values used in the analysis of the mixing ratios are shown in Table 8. The sign of $\delta(\gamma_{3,2})$ is negative as determined by 1970Hr01 and 1974Kr12.

Table 4 – Measured and adopted mixing ratios values for the 112.95005 keV transition.

Reference	$\delta(\gamma_{1,0})$	Comments
1955Ma12		Measured conversion electrons and determined values of K 0.75 (8); L1 0.12 (1); L2 0.70 (7); L3 0.58 (6)
1961We11	-4.0 (2)	Measured conversion electrons and determined values of K 0.78 (8); L 1.40 (14)
1962El02	-5.2 to -4.1	L 1.1 (1) from βγ directional correlation
1964No08	-3.45 (+19-17)	Calculated from L sub-shell ratios, L1/L2: 0.186 (4); L1/L3: 0.204 (4); L2/L3: 1.098 (3)
1968To07	-3.1 (6)	K 0.98 (15) from βγ directional correlation
1970Hr01	-3.7 (3)	From $\gamma\gamma(\theta)$ in ^{177}Lu ($T_{1/2}$ = 6.647 d) β^- decay.
1972Ho39	-4.1 (+5-4)	Calculated from K/Sum(L+M+N+O+P) 0.548 (24); K/L 0.723 (23); L/M 3.97 (3); L/Sum(N+O+P) 14.9 (6); L1/L2 0.170 (3); L1/L3 0.191 (2); L2/L3 1.122 (8); M1/M2 0.152 (6); M1/M3 0.162 (6); M2/M3 1.066 (15); M4,5/M3 0.020 (3)
1972Ho54	-4.75 (7)	From $\gamma\gamma(\theta)$ in ^{177}Lu ($T_{1/2}$ = 6.647 d) β^- decay.
1974Ag01	-4.2 (+9-6)	From $\gamma\gamma(\theta)$ in ^{177}Lu ($T_{1/2}$ = 6.647 d) β^- decay.
1974El02	-4.4 (+5-4)	Calculated from K/L 0.725 (25); L1/L2 0.166 (4); L1/L3 0.187 (3); L2/L3 1.119 (9); L/M 3.982 (33)
1974Kr12	-4.7 (2)	From $\gamma(\theta)$ in ^{177m}Lu ($T_{1/2}$ = 160.44 d) decay.
1977Ke12	-4.8 (2)	From $\gamma(\theta)$ in ^{177}Lu ($T_{1/2}$ = 6.647 d) β^- decay.
1992De53	-4.85 (5)	From $\gamma\gamma(\theta)$ in ^{177}Lu ($T_{1/2}$ = 6.647 d) β^- decay.
2012De24	-4.04 (24)	From ICC ratios in ^{177m}Lu ($T_{1/2}$ = 160.44 d) β^- decay.
Adopted	**-4.67 (12)**	**χ^2/(N-1) = 5.24** (from BrIccMixing)

Table 5 – Measured and adopted mixing ratios values for the 136.7245 keV transition.

Reference	$\delta(\gamma_{2,1})$	Comments
1961We11		Measured conversion electrons and determined values of K 0.71 (7); L 0.67 (7)
1972Gr35	2.9 (4)	
1974Kr12	-3.0 (7)	From $\gamma(\theta)$ in ^{177m}Lu ($T_{1/2}$ = 160.44 d) decay
1974Je02		Measured conversion electrons and determined values of K 0.42 (4); L 0.39 (4)
1974Ag01		Measured conversion electrons and determined values of K 0.49 (8); L1 0.08 (2); L2 0.33 (5); L3 0.30 (5)
2011De07		Measured conversion electrons and determined values of K 0.58 (15); L 0.44 (11); M 0.25 (7)
2012De24	-2.38 (36)	Measured conversion electrons and determined values of K 0.60 (9); M 0.075 (17)
Adopted	**-3.2 (+11-9)**	**χ^2/(N-1) = 3.53** (from BrIccMixing)

Table 6 – Measured and adopted mixing ratios values for the 321.3159 keV transition.

Reference	$\lvert\delta(\gamma_{3,0})\rvert$	Comments
1974Kr12	0.17 (1)	From $\gamma(\theta)$ in ^{177m}Lu ($T_{1/2}$ = 160.44 d) decay
	0.42 (1)	From experimental α_K = 0.087 (3) values reported by 1972Gr35, 1974Ag01, 1974Je02 and 1961We11
	0.42 (1)	From experimental α_L = 0.0169 (8), values reported by 1972Gr35, 1974Ag01, 1974Je02 and 1961We11
Adopted	**0.24 (+4-3)**	**χ^2/(N-1) = 24.6** (from BrIccMixing)

Table 7 – Measured and adopted mixing ratios values for the 208.3661 keV transition.

Reference	$\lvert\delta(\gamma_{3,1})\rvert$	Comments
1973El13		Measured α_{L1}/α_{L2} = 6.2 (5), α_{L2}/α_{L3} = 1.0 (1)
1974Kr12	0.07 (2)	From $\gamma(\theta)$ in ^{177m}Lu ($T_{1/2}$ =160.44 d) decay
1977Ke12	0.08 (2)	From $\gamma(\theta)$ in ^{177}Lu ($T_{1/2}$ = 6.647 d) β^- decay
1961We11	0.07 (3)	From $\gamma\gamma(\theta)$ in ^{177}Lu ($T_{1/2}$ = 6.647 d) β^- decay
2012De24		From ICC ratios in ^{177m}Lu ($T_{1/2}$ = 160.44 d) β^- decay.
Adopted	**0.036 (+15-22)**	**χ^2/(N-1) = 2.79** (from BrIccMixing)

Table 8 – Measured and adopted mixing ratios values for the 71.6425 keV transition.

Reference	$|\delta(\gamma_{3,2})|$	Comments
1970Hr01	-0.017 (7)	From $\gamma\gamma(\theta)$
1974Ag01		Measured α_K = 0.90 (11), α_{L1} = 0.087 (10), α_{L2} = 0.038 (8), α_{L3} = 0.033 (7)
1974Kr12	-0.051 (37)	From $\gamma(\theta)$ in ^{177m}Lu ($T_{1/2}$ = 160.44 d) decay
2011De07		Measured α_K = 0.98 (22), α_L = 0.081 (28), α_M = 0.029 (14)
Adopted	**-0.021 (+13-17)**	**χ^2/(N-1) = 1.5** (from BrIccMixing)

3. Atomic Data

3.1 Hf

The data are from Schönfeld and Janssen (1996Sc06).

3.1.1 X Radiation

While the energies for XKα_2 (Hf) and XKα_1 (Hf) are from Schönfeld and Rodloff (1999ScZX), the XKβ and XL energies are from Firestone (1996FiZX). Relative emission probabilities were calculated using the program EMISSION (2001Sc08).

3.1.2 Auger Electrons

The energies for KLL (Hf), KLX (Hf) and KXY (Hf) are from Schönfeld and Rodloff (1998ScZM). Relative emission probabilities were calculated using the program EMISSION (2001Sc08).

4. Photon Emission

4.1 X-Ray Emission

While the energies for XKα_2 (Hf) and XKα_1 (Hf) are from Schönfeld and Rodloff (1999ScZX), the XKβ and XL energies are from Firestone (1996FiZX). The adopted absolute intensities per 100 disintegrations were calculated using the program EMISSION (2001Sc08). Comparisons between calculated values and the experimental data in 1987Me17 and 2001Sc23, as well as values calculated using the program RADLST (1988Bu*), are presented in Table 9. In general, the agreement between various entries is fairly good, thus suggesting that the ^{177}Lu ground state decay scheme is complete.

4.2 Gamma Emission

The measured relative intensities for transitions following the β^- decay of ^{177}Lu and their adopted values are presented in Table 10. The original values were normalised to I_γ = 100.0 for the 208.3661 keV ($\gamma_{3,1}$) γ-ray.

The measured absolute intensities for the 208.3661 keV ($\gamma_{3,1}$) γ-ray and its corresponding adopted value are presented in Table 11. The latter was used to normalise the relative intensities (Table 10) to absolute values per 100 disintegrations.

Table 9 – Comparison between various Hf X-ray emission intensities per 100 disintegrations.

Emission	Energy (keV)	2025Le18	2016Lu16	2001Sc23	1987Me17	**Adopted**
XLl	6.960	0.0633 (23)		0.0735 (25)	0.087 (5)	**0.0602 (16)**
XLα_2	7.844					
XLα_1	7.899	1.631 (49)		1.51 (3)	1.59 (6)	**1.35 (3)**
XLη	8.139					
XLβ_4	8.905					
XLβ_1	9.023	1.538 (44)		1.34 (3)		
XLβ_6	9.023				1.76 (7)	**1.46 (4)**
XLβ_3	9.163					
XL$\beta_{2,15}$	9.342	0.368 (17)		0.274 (7)		
XLγ_1	10.516	0.288 (11)		0.231 (6)		
XLγ_6	10.733				0.292 (12)	**0.249 (6)**
XLγ_2	10.834	0.0359 (29)		0.0223 (14)		
XLγ_3	10.890					
XL						**3.12 (6)**
XKα_2	54.6120 (7)	1.546 (36)	1.56 (20)	1.55 (3)	1.65 (3)	**1.555 (27)**
XKα_1	55.7909 (8)	2.72 (6)	2.64 (31)	2.73 (6)	2.84 (5)	**2.72 (5)**
XK'β_1	62.985-63.662	0.884 (22)	0.81 (6)	0.885 (15)	0.919 (16)	**0.898 (21)**
XK'β_2	64.942-65.316	0.284 (6)	0.215 (24)	0.238 (5)	0.252 (5)	**0.240 (7)**
XKβ						**1.136 (22)**

Table 10 – Relative γ-ray intensities for transitions following β^- decay of ^{177}Lu.

Gamma	$\gamma_{1,0}$	$\gamma_{2,1}$	$\gamma_{2,0}$	$\gamma_{3,2}$	$\gamma_{3,1}$	$\gamma_{3,0}$
Energy (keV)	112.95005 (36)	136.7245 (5)	249.6742 (6)	71.6425 (6)	208.3661 (2)	321.3159 (6)
2025Le18	59.49 (44)	0.450 (8)	1.923 (14)	1.635 (22)	100.0	1.972 (37)
2016Dr15	60.0 (5)				100.0	
2016Lu16	55 (5) @	0.32 (2) @	1.74 (5) @	1.52 (11)	100.0	1.89 (19)
2012Ko24	59.0 (7)	0.466 (12)	1.918 (22)	1.63 (2)	100.0	2.00 (2)
2011De07	62.15 (5) #	0.543 (6) #	1.982 (4) ‡	1.780 (7) #	100.0	2.470 (7) @
2010Di18	60.1 (5)				100.0	
2001Sc23	59.6 (6)	0.448 (8)	1.918 (17)	1.674 (21)	100.0	2.002 (19)
1987Me17	59.6 (8)	0.457 (6)	2.00 (2) †	1.71 (5)	100.0	2.17 (2) *
1974Ag01	60 (5)	0.52 (5) @	1.9 (2)	1.5 (1)	100.0	1.98 (18)
1967Ha09	61 (4) @	0.56 (4) @	1.83 (12) @	1.46 (6) #	100.0	2.20 (12)
1964Al04	58 (4) @	0.43 (3)	1.93 (14)	1.40 (10) #	100.0	1.99 (14)
1961We11	62 (2) @	0.47 (15)	2.00 (20)	0.30 (10) #	100.0	2.28 (10)
1955Ma12	45.5 #		1.36 #	0.91 #	100.0	1.45 (29) @
Adopted	**59.69 (23)**	**0.4538 (39)**	**1.925 (9)**	**1.646 (12)**	**100.0**	**2.011 (12)**
χ^2 / χ^2_{crit}	**0.38 / 2.80**	**0.54 / 3.02**	**0.66 / 2.80**	**1.55 / 3.02**		**2.35 / 2.51**

@ Data rejected owing to the Chauvenet criterion. † Uncertainty doubled. * Uncertainty tripled.
# Value not used in the analysis. ‡ Uncertainty expanded to that of 2012Ko24.

Table 11 – Absolute emission probabilities per 100 disintegrations for the 208.3661 keV γ-ray.

Reference	Absolute Intensity for $\gamma_{3,1}$ (%)	Comments
2025Le18	10.48 (7)	
2016Dr15	10.38 (9)	
2016Lu16	10.80 (35)	
2012Ko24	10.55 (9)	
2010Di18	10.35 (8)	
2001Sc23	10.36 (7)	
1964Cr02	10.7 (5)	
1961We11	11.4 (6)	Excluded by the Chauvenet criterion
Adopted	**10.425 (35)**	
χ^2 / χ^2_{crit}	**1.00 / 2.80**	

5. Electron Emission

The electron energies and emission probabilities were calculated using EMISSION (v3.10, 28-Jan-2003) described in 2000Sc47. The β^- transition endpoint energies were determined using Q(β^-) = 496.8 (8) keV (2021Wa16) and the individual level energies, which were determined using the GTOL code based on a fit to the recommended γ-ray energies. The mean β^- energies were calculated using the BetaShape program (2023Mo21). The adopted values for the β^- transition emission probabilities were determined from the total (photons + conversion electrons) γ-ray emission probability balances at each level.

6. References

1949Do05 D.G. Douglas, Phys. Rev. 75, 1960 (1949) [P_{β^-}]

1955Ma12 P. Marmier, F. Boehm, Phys. Rev. 97, 103 (1955) [P_{β^-}, E_γ, I_γ]

1956Wi39 T. Wiedling, Thesis, Univ. Stockholm (1956) [P_{β^-}]

1958Be41 R.H. Betts, O.F. Dahlinger, D.M. Munro, Can. J. Phys. 36, 73 (1958) [$T_{1/2}$]

1960Sc19 L.C. Schmid, W.P. Stinson, Nucl. Sci. Eng. 7, 477 (1960) [$T_{1/2}$]

1961Ha21 O. Hansen, M.C. Olesen, O. Skilbreid, B. Elbek, Nucl. Phys. 25, 634 (1961) [$T_{1/2}$]

1961Ha38 U. Hauser, K. Runge, G. Knissel, Nucl. Phys. 27, 632 (1961) [$T_{1/2}$]

1961We11 H.I. West, Jr., L.G. Mann, R.J. Nagle, Phys. Rev. 124, 527 (1961) [E_γ, I_γ, δ, ICC]

1962Be46 E.E. Berlovich, Y.K. Gusev, V.V. Ilin, M.K. Nikitin, Zhur. Eksptl. i Teoret. Fiz. 43, 1625 (1962); Soviet Phys. JETP 16, 1144 (1963) [$T_{1/2}$]

1962El02 M.S. El-Nesr, E. Bashandy, Nucl. Phys. 31, 128 (1962) [P_{β^-}]

1964Al04 P. Alexander, F. Boehm, E. Kankeleit, Phys. Rev. 133, B284 (1964) [P_{β^-}, E_γ, I_γ]

1964Cr02 D.F. Crouch, L.D. McIsaac, IDO-16932, p.26 (1964) [I_γ]

1964No08 T. Novakov, J.M. Hollander, Nucl. Phys. 60, 593 (1964) [δ]

1965Ma18 B.P.K. Maier, Z. Physik 184, 153 (1965) [E_γ]

1965Ro17 R. Rougny, J.J. Samueli, A. Sarazin, J. Phys. (Paris) 26, 63 (1965) [$T_{1/2}$]

1967Ha09 A.J. Haverfield, F.M. Bernthal, J.M. Hollander, Nucl. Phys. A94, 337 (1967) [P_{β^-}, E_γ, I_γ]

1968Ra25 B.V.N. Rao, S. Jnanananda, Indian J. Pure Appl. Phys. 6, 555 (1968) [$T_{1/2}$]

1968To07 S. Tornkvist, S. Strom, J.E. Thun, V. Schmidt, N. Baade, Nucl. Phys. A 117, 336 (1968) [δ]
1970Hr01 B. Hrastnik, I. Basar, M. Diksic, K. Ilakovac, V. Kos, A. Ljubicic, Z. Phys. 239, 25 (1970) [δ]
1972Da35 L.M. Dautov, Y.A. Lysikov, U.M. Makhanov, Y.K. Shubnyi, Izv. Akad. Nauk SSSR, Ser. Fiz. 36, 2544 (1972); Bull. Acad. Sci. USSR, Phys. Ser. 36, 2210 (1973) [$T_{1/2}$]
1972Em01 J.F. Emery, S.A. Reynolds, E.I. Wyatt, G.I. Gleason, Nucl. Sci. Eng. 48, 319 (1972) [$T_{1/2}$]
1972Gr35 V.N. Grigorev, D.M. Kaminker, Y.V. Sergeenkov, Izv. Akad. Nauk SSSR, Ser. Fiz. 36, 842 (1972); Bull. Acad. Sci. USSR, Phys. Ser. 36, 762 (1973) [ICC]
1972Ho39 S. Hogberg, R. Jadrny, S.E. Karlsson, G. Malmsten, O. Nilsson, Z. Phys. 254, 89 (1972) [δ]
1972Ho54 L. Holmberg, V. Stefansson, J. Becker, C. Bargholtz, L. Gidefeldt, Phys. Scr. 6, 177 (1972) [δ]
1973El13 M.S. El-Nesr, E. Bashandy, Z. Naturforsch. 28a, 1959 (1973) [δ]
1973GoYH K.P. Gopinathan, A.P. Agnihotry, S.B. Patel, M.S. Bidarkundi, Proc. Int. Conf. Nuclear Moments and Nuclear Structure, Osaka, Japan (1972), H. Horie, K. Sugimoto, Eds., p.430 (1973); J. Phys. Soc. Jap. 34 Suppl. (1973) [$T_{1/2}$]
1974Ag01 A.P. Agnihotry, K.P. Gopinathan, H.C. Jain, Phys. Rev. C9, 336 (1974) [I_γ, ICC, δ]
1974El02 M.S. El-Nesr, E. Bashandy, M.G. Mousa, Z. Naturforsch. 29a, 593 (1974) [δ]
1974Je02 B.D. Jeltema, F.M. Bernthal, Phys. Rev. C10, 778 (1974) [ICC]
1974Kr12 K.S. Krane, C.E. Olsen, W.A. Steyert, Phys. Rev. C10, 825 (1974) [$\gamma(\theta)$]
1977Ke12 H.E. Keus, W.J. Huiskamp, Physica 85B, 137 (1977) [δ]
1981Hn03 V. Hnatowicz, Czech. J. Phys. B31, 260 (1981) [E_γ, I_γ]
1982La25 F. Lagoutine, J. Legrand, Int. J. Appl. Radiat. Isotop. 33, 711 (1982) [$T_{1/2}$]
1987Me17 D. Mehta, B. Chand, S. Singh, M.L. Garg, N. Singh, T.S. Cheema, P.N. Trehan, Nucl. Instrum. Methods Phys. Res. A260, 157 (1987) [I_γ]
1988Bu* T.W. Burrows, Report BNL-NCS-52142, February 29, 1988 [RADLST program]
1988WoZO M.J. Woods and A.S. Munster, National Physics Laboratory, Teddington, UK, Rep. RS(EXT) 95, (1988) [LWEIGHT program]
1989Ma56 S. Matsui, H. Inoue, Y. Yoshizawa, Nucl. Instrum. Methods Phys. Res. A281, 568 (1989) [E_γ]
1990Ab02 A. Abzouzi, M.S. Antony, A. Hachem, V.B. Ndocko Ndongue, J. Radioanal. Nucl. Chem. 144, 359 (1990) [$T_{1/2}$]
1992De53 C.C. Dey, B.K. Sinha, R. Bhattacharya, Nuovo Cim. 105A, 1307 (1992) [$T_{1/2}$, δ]
1996Al20 I. Alfter, E. Bodenstedt, W. Knichel, J. Schuth, R. Vianden, Z. Phys. A 355, 363 (1996) [$T_{1/2}$]
1996FiZX R.B. Firestone, Table of Isotopes, 8th Ed., V.S. Shirley, C.M. Baglin, S.Y.F. Chu, J. Zipkin Eds., John Wiley and Sons, Inc., New York, Vol.2 (1996) [Nuclear data]
1996Sc06 E. Schönfeld, H. Janssen, Nucl. Instrum. and Methods A369, 527 (1996) [Atomic data]
1998ScZM E. Schönfeld, G. Rodloff, PTB-6.11-98-1 (1998) [KLL, KLX, KXY energies]
1999ScZX E. Schönfeld, G. Rodloff, PTB-6.11-1999-1 (1999) [X-ray energies]
2000Sc47 E. Schönfeld, H. Janssen, Appl. Radiat. Isot. 52 (2000) 595 [X-ray and Auger Electron emission probabilities]
2001Sc08 E. Schönfeld, U. Schötzig, Appl. Radiat. Isot. 54, 785 (2001) [Emission probabilities]

2001Sc23 U. Schötzig, H. Schrader, E. Schönfeld, E. Gunther, R. Klein, Appl. Radiat. Isot. 55, 89 (2001) [P_{β^-}, $T_{1/2}$, I_γ]

2001Zi01 B.E. Zimmerman, M.P. Unterweger, J.W. Brodack, Appl. Radiat. Isot. 54, 623 (2001) [$T_{1/2}$]

2002Ba85 I.M. Band, M.B. Trzhaskovskaya, C.W. Nestor Jr., P.O. Tikkanen, S. Raman, At. Data Nucl. Data Tables 81, 1 (2002) [ICCs]

2002Un02 M.P. Unterweger, Appl. Radiat. Isot. 56, 125 (2002) [$T_{1/2}$]

2003Ko33 F.G. Kondev, Nucl. Data Sheets 98, 801 (2003) [Decay scheme evaluation]

2004BeZQ M.M. Bé, V. Chisté, C. Dulieu, E. Browne, V. Chechev, N. Kuzmenko, R. Helmer, A.L. Nichols, E. Schönfeld, R. Dersch, Monographie BIPM-5, Vol.2, Bureau International des Poids et Mesures (2004) [Decay scheme evaluation]

2004Sc04 H. Schrader, Appl. Radiat. Isot. 60, 317 (2004) [$T_{1/2}$]

2008Ca13 L. Canella *et al.*, J. Radioanal. Nucl. Chem. 276, 813 (2008) [$T_{1/2}$]

2008DuZX C. Dulieu, M.M. Bé, V. Chisté, Proc. Int. Conf. on Nuclear Data for Science and Technology, 22-27 April 2007, Nice, France (2008) 97 [SAISINUC software]

2008Ki07 T. Kibédi, T.W. Burrows, M.B. Trzhaskovskaya, P.M. Davidson, C.W. Nestor Jr, Nucl. Instrum. Methods Phys. Res. A589 (2008) 202 [Theoretical ICCs]

2010Di18 M.S. Dias, F.F.V. Silva, M.F. Koskinas, Appl. Radiat. Isot. 68, 1349 (2010) [I_γ]

2011De07 S. Deepa, K. Vijay Sai, R. Gowrishankar, D.R. Rao, K. Venkataramaniah, Appl. Radiat. Isot. 69, 869 (2011) [ICC, E_γ, P_γ]

2011Zh56 S. Zhu *et al.*, Nucl. Instrum. Methods Phys. Res. A 652, 231 (2011) [$T_{1/2}$]

2011Po07 S. Pommé, J. Paepen, T. Altzitzoglou, R. Van Ammel, E. Yeltepe, Appl. Radiat. Isot. 69, 1267 (2011) [$T_{1/2}$]

2012De24 S. Deepa, K. Vijay Sai, R. Gowrishankar, K. Venkataramaniah, Eur. Phys. J. A 48, 126 (2012) [ICC, E_γ]

2012Fi12 R. Fitzgerald, J. Res. Natl. Inst. Stand. Technol. 117, 80 (2012) [$T_{1/2}$]

2012Ko24 K. Kossert, O.J. Nähle, O. Ott, R. Dersch, Appl. Radiat. Isot. 70, 2215 (2012) [$T_{1/2}$, P_γ]

2012Re25 E.A. Rezende, A.R. Correia, A. Iwahara, C.J. da Silva, L. Tauhata, R. Poledna, R.L. da Silva, E.M. de Oliveira, A.E. de Oliveira, Appl. Radiat. Isot. 70, 2081 (2012) [$T_{1/2}$]

2013Do24 N. D'Olympia, S. Lakshmi, P. Chowdhury, E.G. Jackson, J. Glodo, K. Shah, Nucl. Instrum. Methods Phys. Res. A 728, 31 (2013) [$T_{1/2}$]

2015Ki* T. Kibédi, unpublished, available from http://www.nndc.bnl.gov/nndcscr/ensdf_pgm/analysis/BrIcc/ [δ]

2016Dr15 P. Dryàk, J. Sochorová, J. Šolc, P. Auerbach, Appl. Radiat. Isot. 109, 160 (2016) [$T_{1/2}$]

2016Ke06 M.A. Kellett, Appl. Radiat. Isot. 109, 129 (2016) [Decay scheme evaluation]

2016Lu16 A. Luca, M. Sahagia, M.-R. Ioan, A. Antohe, B. Luminita Savu, Appl. Radiat. Isot. 109, 146 (2016) [$T_{1/2}$]

2017FeZZ K.M. Ferreira, S.M. Collins, A.J. Fenwick, EPJ Web Conf. 146, 08002 (2017) [$T_{1/2}$]

2019Ko26 F.G. Kondev, Nucl. Data Sheets 159, 1 (2019) [ENSDF evaluation]

2020Kn04 L. Knafla *et al.*, Phys. Rev. C 102, 054322 (2020) [$T_{1/2}$]

2021Wa16 M. Wang, W.J. Huang, F.G. Kondev, G. Audi, S. Naimi, Chin. Phys. C 45, 030003 (2021) [Q-value]

2023Ma26 J. Marganiec-Gałązka, M. Czudek, E. Lech, A. Listkowska, P. Saganowski, Z. Tymiński, T. Ziemek, Appl. Radiat. Isot. 197, 110829 (2023) [$T_{1/2}$]

2023Mo21 X. Mougeot, Appl. Radiat. Isot. 201, 111018 (2023) [BetaShape code]

2025Le18 M.-C. Lépy, B. Boyer, S. Pierre, A. Utepov, C. Bobin, L. Chambon, Appl. Radiat. Isot. 226, 112212 (2025) [I_γ]